

Analyzing PZ-SIC for reaction barrier heights beyond the LDSA: Unraveling the Evolution of Stretched Bond Errors

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Self-interaction Error (SIE)

□ Density functional theory (DFT) total energy:

$$E^{\text{DFT}}[n_{\uparrow}, n_{\downarrow}] = \underbrace{T + E_{\text{ext}}[n] + E_{\text{H}}[n]}_{\text{Known}} + \underbrace{E_{\text{xc}}[n_{\uparrow}, n_{\downarrow}]}_{\text{Unknown} \rightarrow \text{Approximate}}$$

□ For a 1-e density n_i , the exact (yet unknown) functional satisfies,

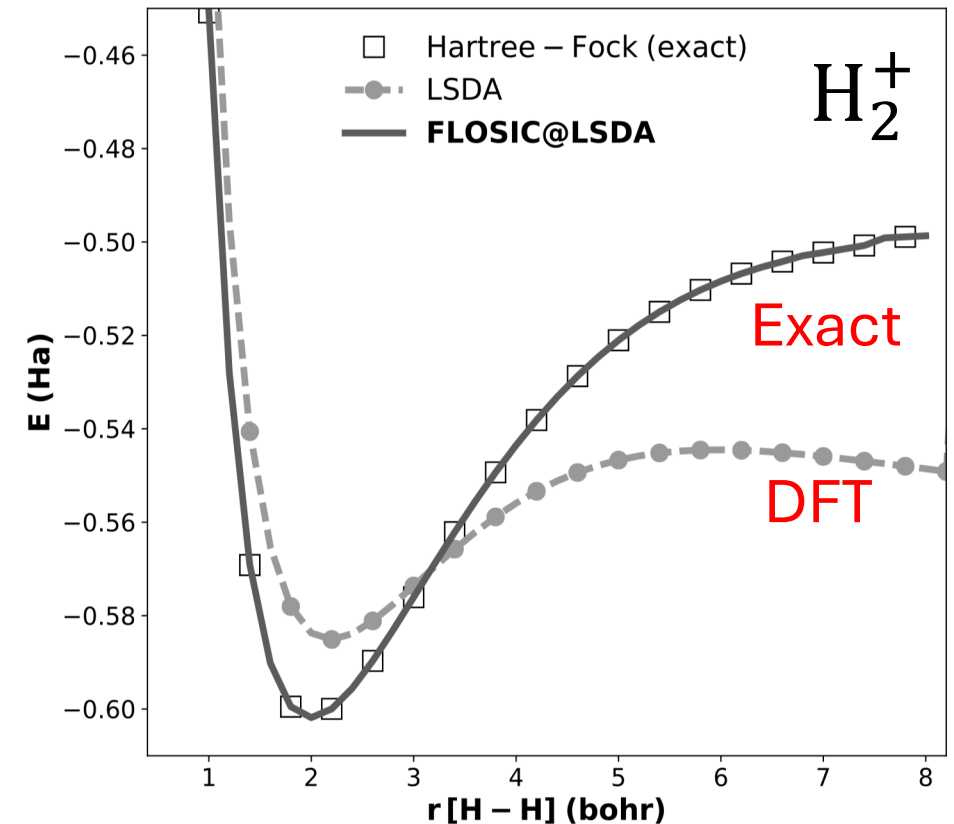
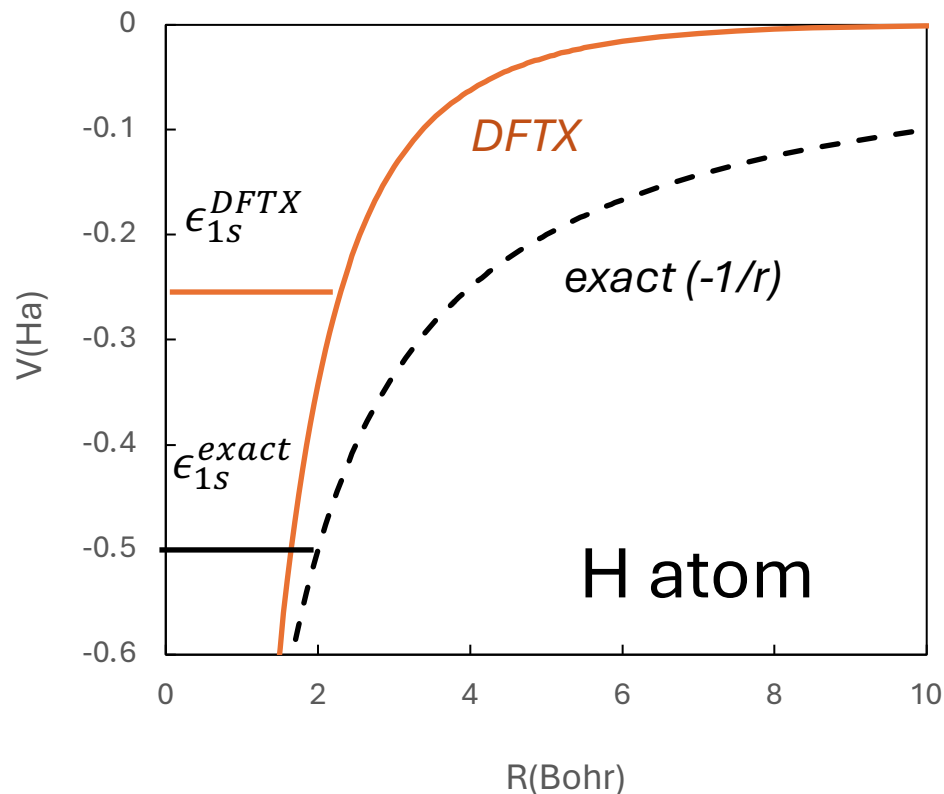
$$E_{\text{H}}[n_i] = -E_{\text{xc}}[n_i, 0]$$

so that $E_{\text{H}} + E_{\text{xc}}$ nullify exactly, eliminating any **self-interaction**

□ Approximate density functionals fail to meet this criterion, resulting in a leftover and unrealistic self-interaction

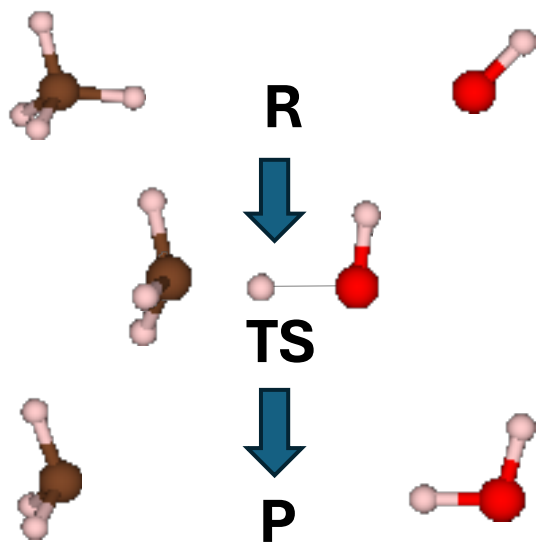
The consequences of SIE

- Formally: DFAs not exact in the 1-e limit
- In practice: poor orbital energies; unbound anions; incorrect charge transfer; unphysical dissociation; **poor reaction barriers**; etc.

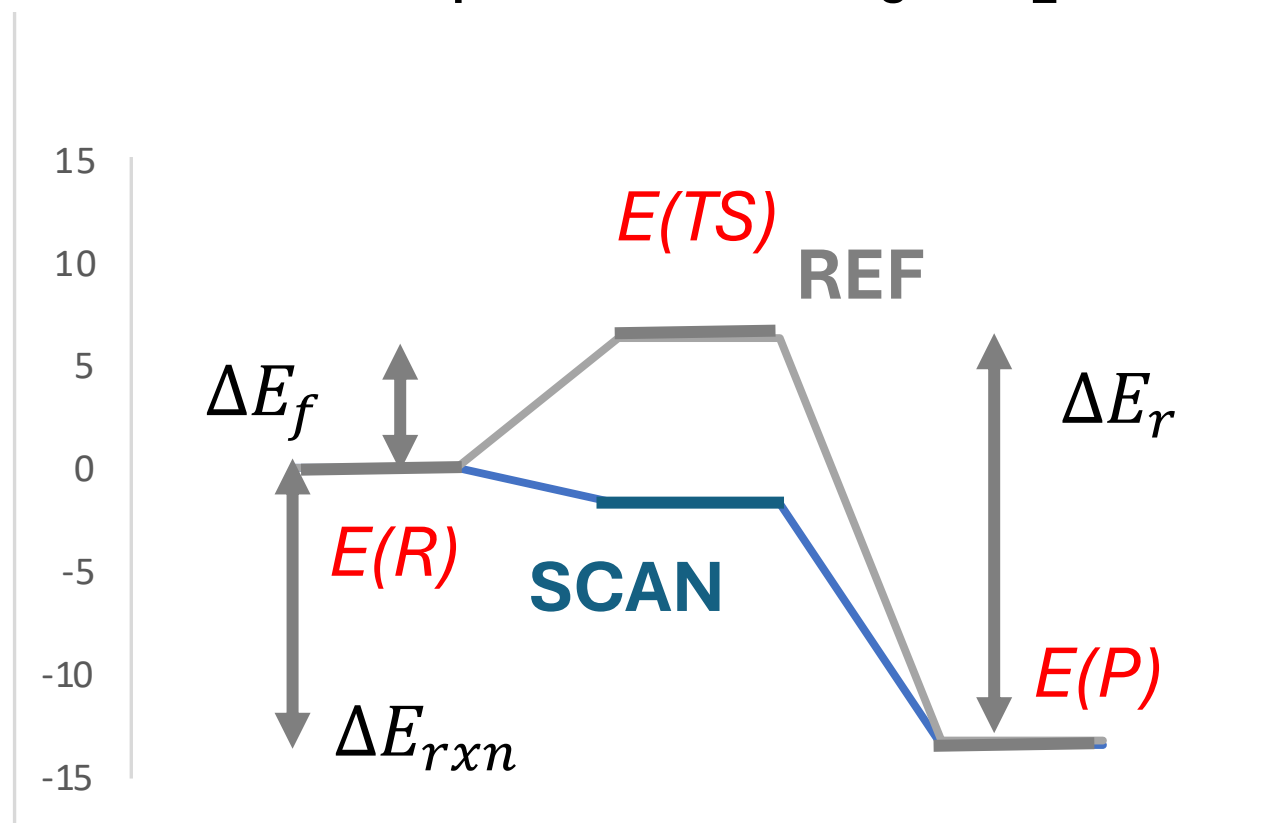


Chemical Reaction Barriers without SIC

Underestimating reaction barriers an infamous problem for DFAs.



BH6 Set



SIC: removing self-interaction from DFA

□ Orbital-by-orbital removal of self-interaction: Perdew-Zunger¹ SIC

$$\begin{aligned} E^{\text{DFA-SIC}} &= E^{\text{DFA}} + \sum_{i\sigma} \left\{ -\frac{1}{2} E_H(n_{i\sigma}) - E_{xc}^{\text{DFA}}[n_{i\sigma}; 0] \right\} \\ &= E^{\text{DFA}} + \sum_{i\sigma} E^{\text{SIC}}[n_{i\sigma}] \quad (n(\vec{r}) = \sum_{i,\sigma} n_{i\sigma}(\vec{r}) \rightarrow \\ &\quad \text{orbital dependent } n_{i\sigma}?) \end{aligned}$$

□ Conditions for minimum energy:²

$$H^{\text{DFA-SIC}} |\phi_{i\sigma}\rangle = (H_{o\sigma} + V_{i\sigma}^{\text{SIC}}) |\phi_{i\sigma}\rangle = \sum_j \lambda_{ji}^{\sigma} |\phi_{j\sigma}\rangle$$

$$\langle \phi_{j\sigma} | V_{j\sigma}^{\text{SIC}} - V_{i\sigma}^{\text{SIC}} | \phi_{i\sigma} \rangle = 0, \quad \forall i, j \text{ pairs}$$

- ✓ “Localization equations”
- ✓ Order N^2 constraints
- ✓ Localized orbitals maximize |self-Hartree|

¹Perdew and Zunger, PRB **23**, 5048 (1981)

²Pederson, Heaton, and Lin, JCP **82**, 2688 (1985)

Fermi-Löwdin orbital SIC (FLOSIC)

□ Fermi orbitals (F_i):

$$F_i(\vec{r}) = \frac{\sum_{\alpha} \psi_{\alpha}^* (\vec{a}_i) \psi_{\alpha}(\vec{r})}{n(r)}$$

□ $\vec{a}_i$: Fermi orbital descriptors (FODs)

□ F_i are normalized, not orthonormal $\rightarrow$ Löwdin orthogonalization

□ Fermi-Löwdin orbitals (FLOs) fully specified by ψ_{α} and $\vec{a}_i$

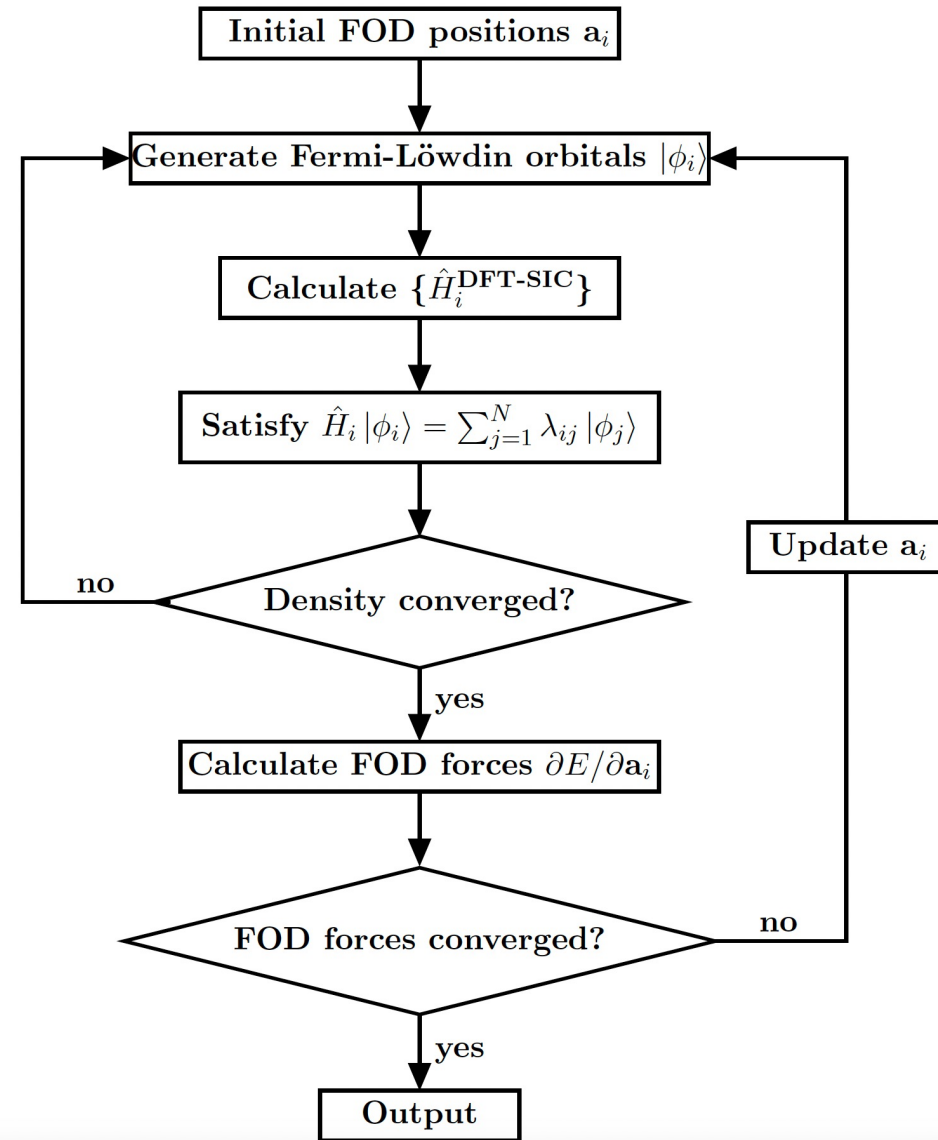
□ Variationally finding $\vec{a}_i$ that minimizes the energy

□ $\frac{\partial E}{\partial \vec{a}_i}$ can be computed² $\rightarrow$ gradient-based FOD optimization.

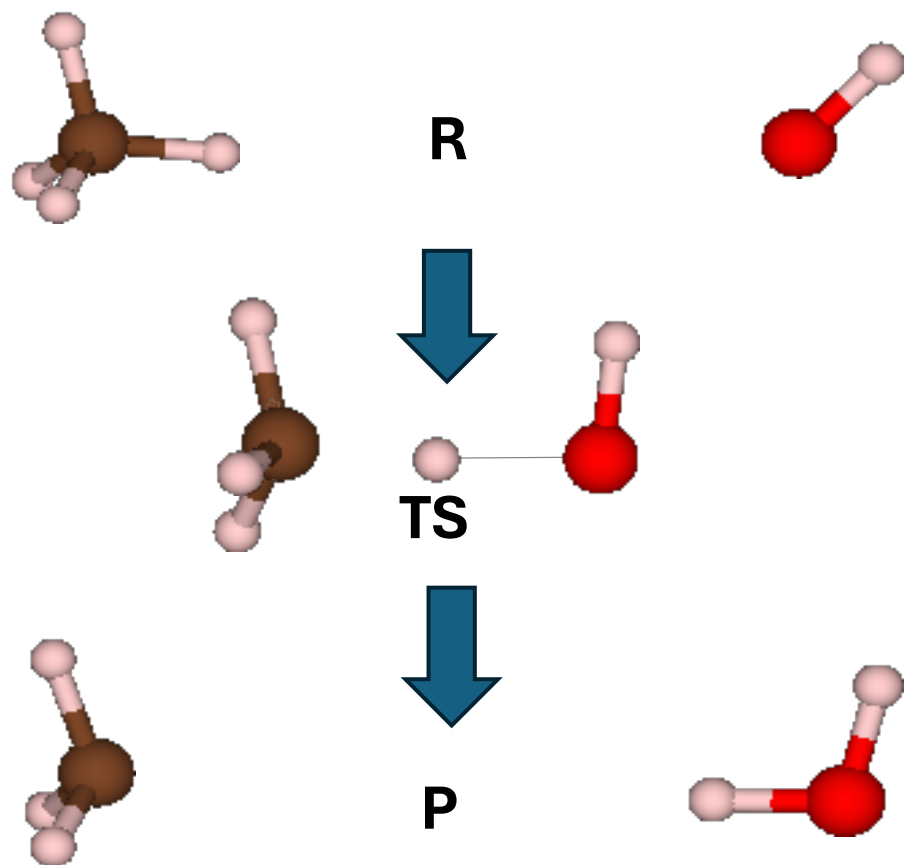
¹Pederson, Perdew, Ruzsinszky, JCP **140**, 121103 (2014),

²Pederson, JCP **142**, 064112 (2015)

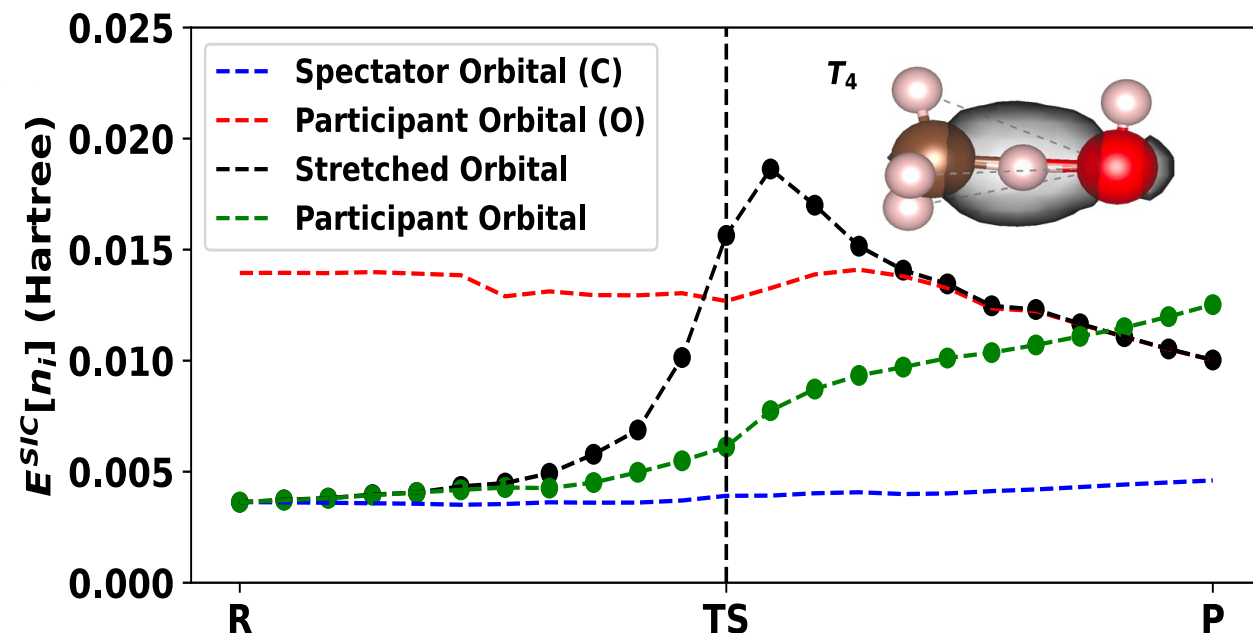
³K. A. Jackson *et al* 2019 *J. Phys.: Conf. Ser.* **1290** 012002



Evolution of SIC along reaction path



U^{SIC} vs approx. rxn path



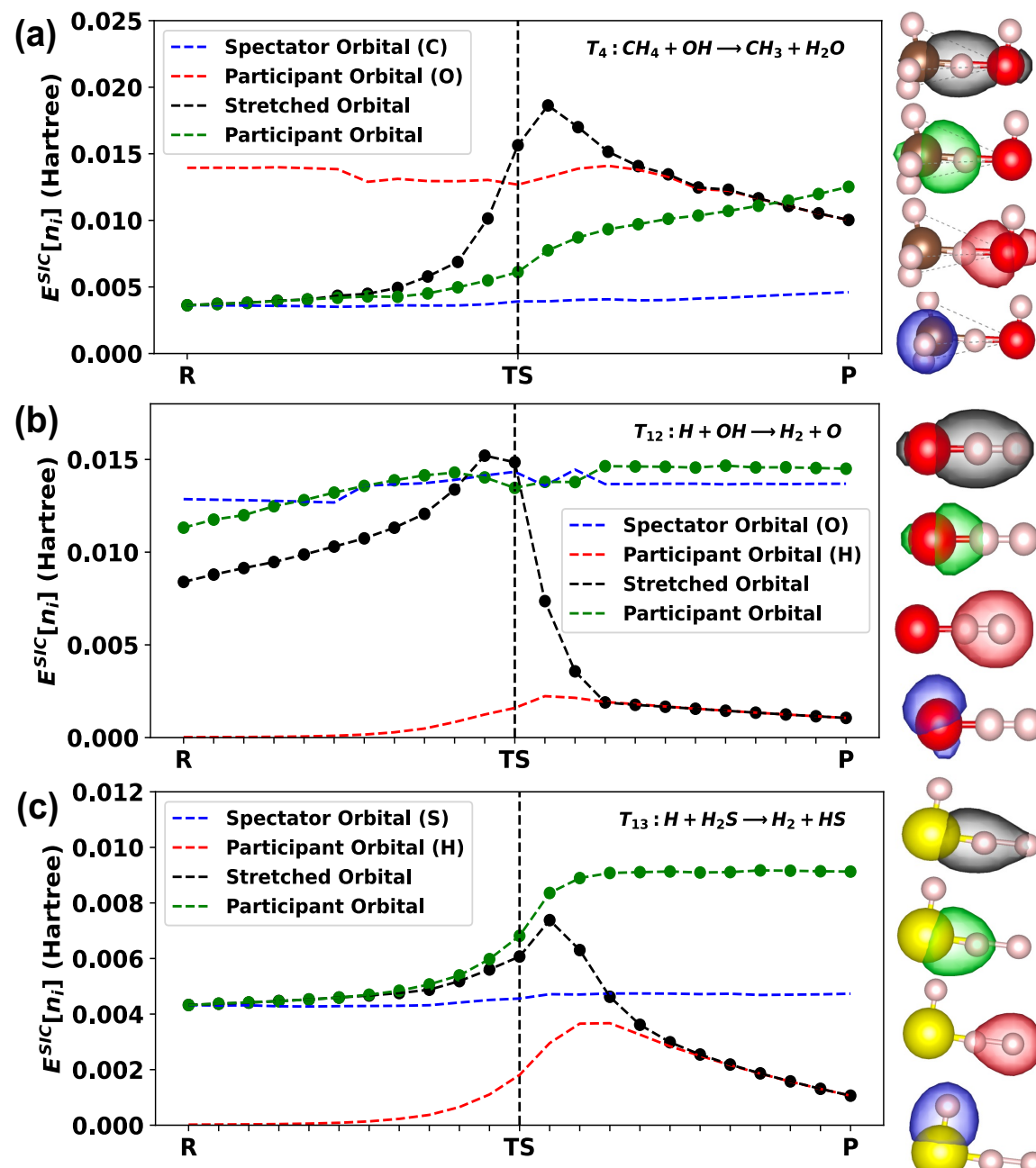
- ❑ Changes in E^{SIC} more pronounced for participant than spectator orbitals
- ❑ Stretched bond emerges near TS

Evolution, cont'd

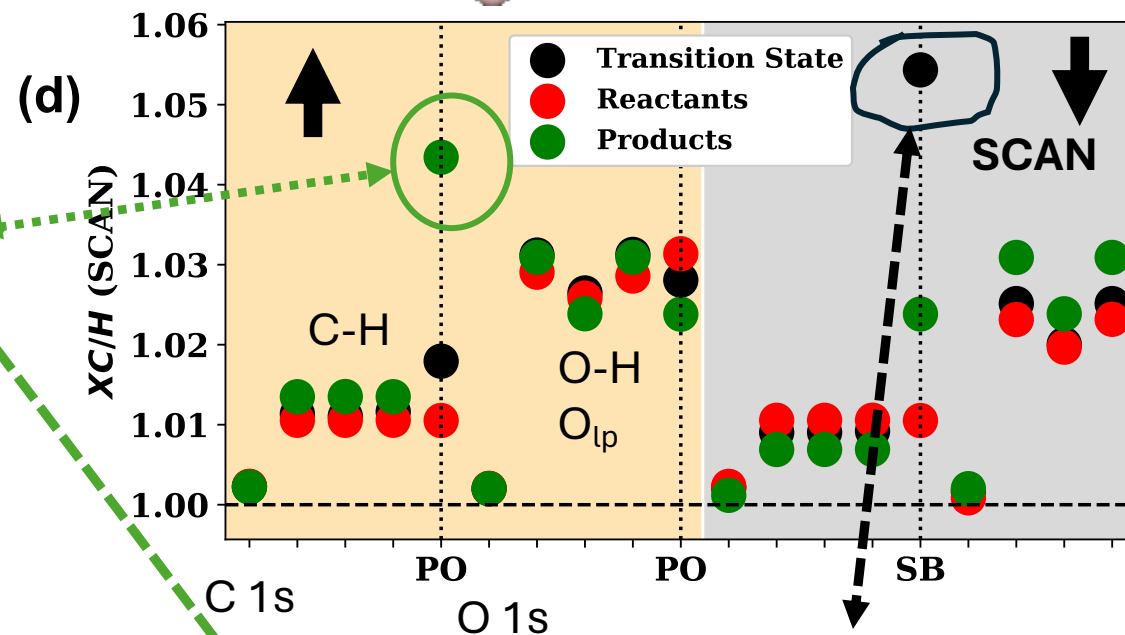
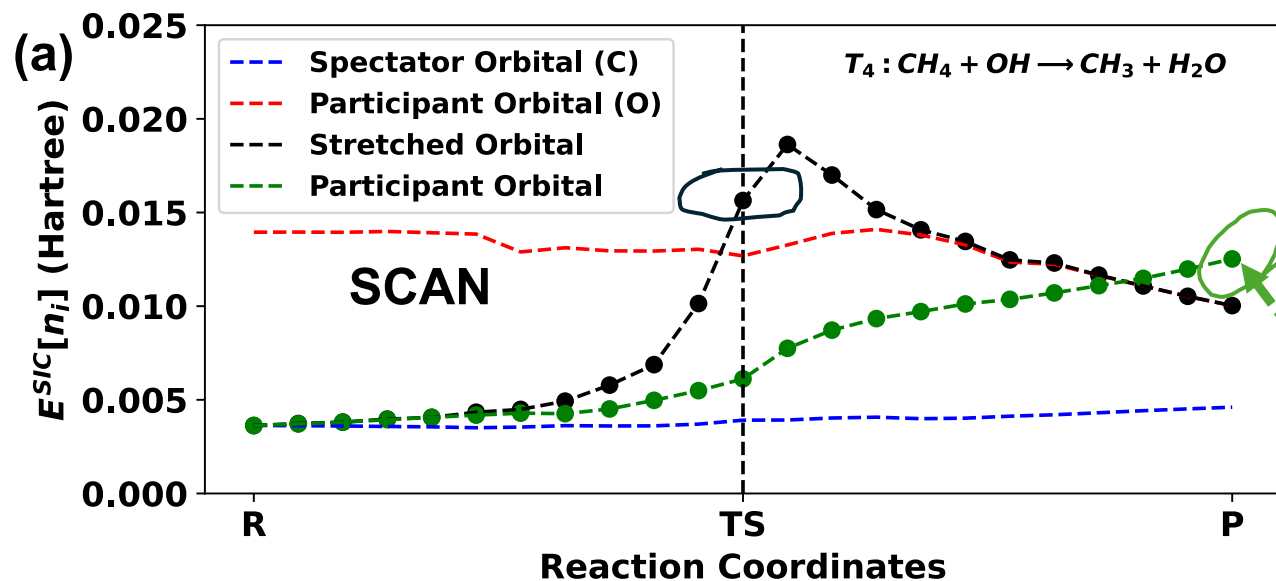
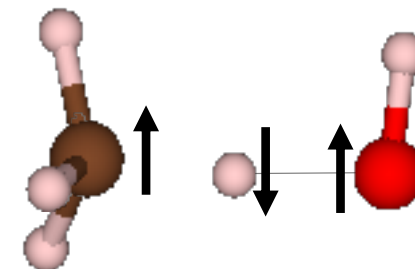
Same features in each reaction:

- ❑ Spectator orbitals ~unchanged
- ❑ Participant orbitals evolve
- ❑ E^{SIC} peaks near TS geometry for stretched bond

Singh, Y., Peralta, J. E., & Jackson, K. A. (2024). JCP, 160(12), 124105.

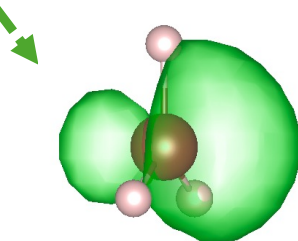


Identifying stretched orbitals: XC/H

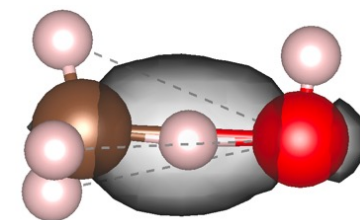


Larger $XC/H \rightarrow$ more positive E^{SIC}

- ☐ Stretched-bond
- ☐ CH₃ radical orbital 2p-like

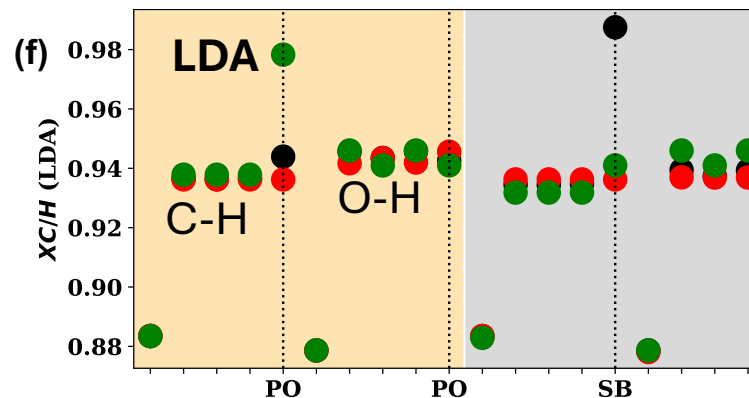
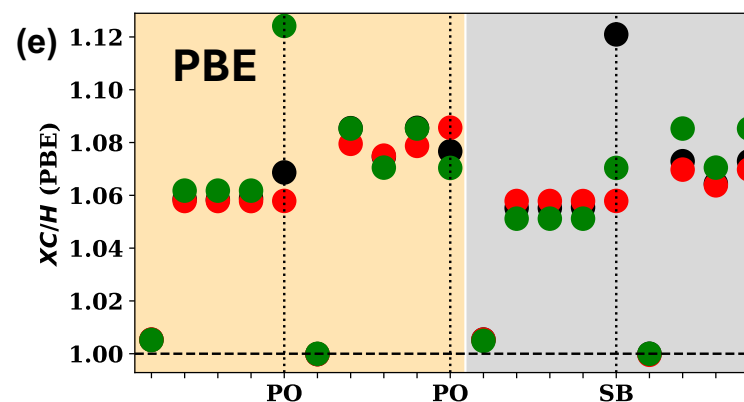
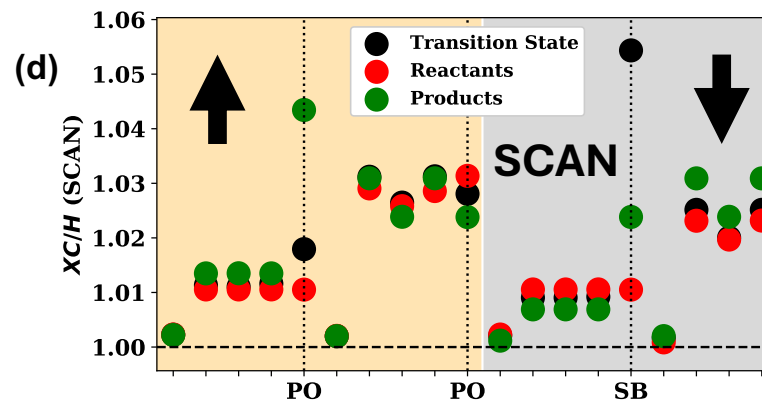
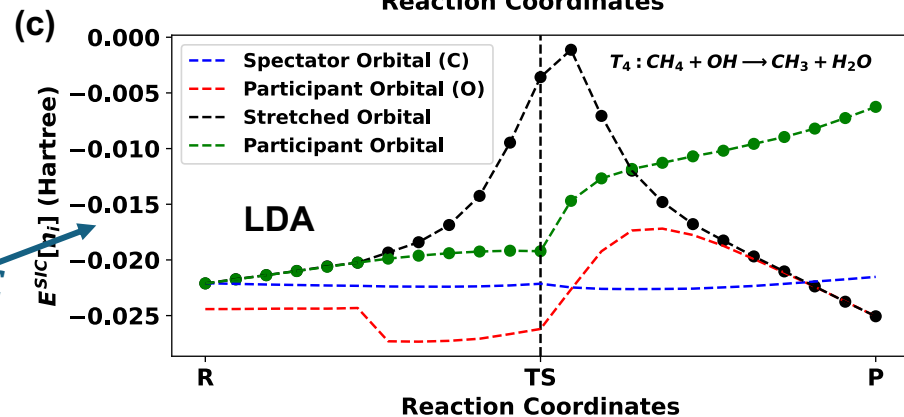
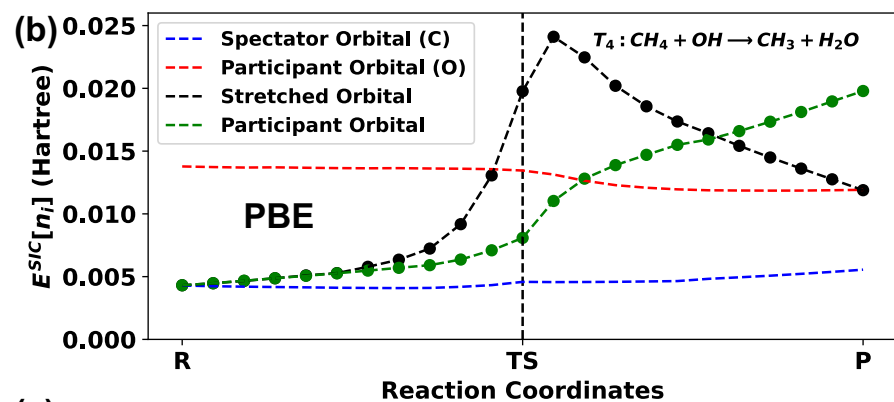
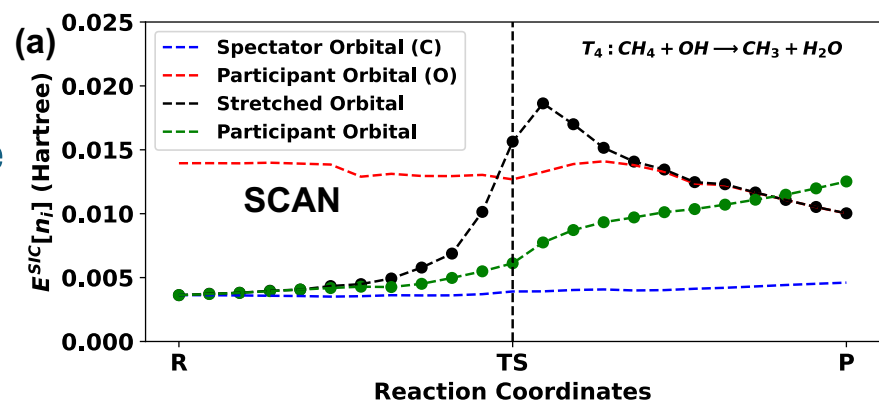


CH₃ radical



Stretched

$U^{SIC}, XC/H$



☐ Stretched bond story same for all DFAs

☐ XC/H shifts similar; values vary

☐ $XC/H \rightarrow$ sign of E^{SIC}

☐ Smallest E^{SIC} for FLO-SCAN

SIC and Density-driven Error

BH6 error statistics (kcal/mol)*

Functionals	ΔE_f^{MAD}	ΔE_r^{MAD}	ΔE_{tot}^{MAD}	ΔE_{rxn}^{MAD}
LDA	15.62	20.55	18.08	9.19
LDA@FLO	12.76	16.89	14.83	8.44
FLO-LDA	1.94	8.25	5.10 (5.02)	6.31 (5.56)
PBE	8.08	11.14	9.61	3.90
PBE@FLO	4.97	7.67	6.32	3.42
FLO-PBE	3.06	5.23	4.14 (3.37)	6.94 (6.74)
SCAN	7.43	8.32	7.88	1.34
SCAN@FLO	6.21	6.99	6.60	1.26
FLO-SCAN	2.54	3.64	3.09 (3.04)	4.70 (4.80)
B3LYP ²⁵	5.10	4.56	4.83	1.67
LC- ω PBE ²⁵	0.31	2.10	1.21	1.81
SX50 ²⁵	3.23	1.17	2.20	2.88

 barriers
  rxn energy

- Slight (1.3 kcal) improvement of SCAN BH6 *barriers* from density.
- Larger improvement from ΔE^{SIC} (3.5 kcal).
- No density improvement of rxn energy.
- *FLO-SCAN correction due to functional, not density*

Summary

- ❑ BH6 $\subset$ BH76 reaction barriers significantly improved in FLO-DFA
- ❑ SIC contributions of PO most important for BH6 reactions; stretched bond orbital contribution largest of all.
- ❑ XC/H “lobedness” indicator $\rightarrow$ “size” of U^{SIC}
- ❑ FLO-DFA barriers *improved by change in functional*, not change in density

Acknowledgments



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- Ms. Priyanka Shukla
- Prof. Karl Johnson



Thank You



SIC and density-driven error

	F/R	Ref	@LC ω PBE	@SCAN-50	@FLO	@HF	SCAN
T4: CH ₄ + OH $\rightarrow$ CH ₃ + H ₂ O	F	6.3	-7.26	-6.5	-6.03	-1.86	-7.70
	R	19.5	-7.41	-6.6	-6.23	-1.86	-7.72
T12: H + OH $\rightarrow$ O + H ₂	F	10.9	-7.39	-6.75	-6.37	-2.81	-7.68
	R	13.2	-10.53	-9.83	-9.29	-5.41	-10.93
T13: H + H ₂ S $\rightarrow$ H ₂ + HS	F	3.9	-6.17	-6.27	-6.15	-4.6	-6.68
	R	17.2	-4.93	-5.06	-5.5	-2.99	-5.62

SCAN@X Barrier errors (kcal/mol)

@X: evaluated on self-consistent X density

- ❑ Kaplan *et al.*: LC ω PBE good “proxy” for exact density¹
- ❑ @LC ω PBE, @FLO-SCAN change SCAN barrier by similar small amounts
- ❑ HF density changes SCAN barrier much more $\rightarrow$ error cancellation^{1,2}
- ❑ FLO barrier improvement due to PZSIC functional

¹Kaplan *et al.*, J. Chem Theor. Comp. **19**, 532 (2023); ²Kanungo *et al.*, J. Phys. Chem. Lett. **15**, 323 (2024)

Reactions $\rightarrow$	T_4		T_{12}		T_{13}		T_9	
	ΔE_f	ΔE_r	ΔE_f	ΔE_r	ΔE_f	ΔE_r	ΔE_f	ΔE_r
LDA	-23.71	-17.32	-12.51	-27.28	-10.64	-17.05	-25.32	-8.58
LDA@FLO	-18.96	-12.50	-9.99	-22.54	-9.33	-15.64	-16.88	-0.79
LDA@FLO-SCAN	-18.54	-12.14	-10.13	-22.03	-9.37	-15.53	-15.82	-0.11
FLO-LDA	-1.90	-12.49	-1.28	-4.74	-2.64	-7.52	-4.55	-7.46
ΔE^{SIC}	17.06	0.01	8.71	17.80	6.69	8.12	12.33	-6.67
ΔE_{SB}^{SIC}	11.62	13.48	13.12	10.64	3.47	6.06	8.63	10.70
PBE	-11.81	-10.55	-7.30	-14.91	-5.13	-7.94	-14.57	-9.04
PBE@FLO	-7.17	-6.10	-4.00	-10.72	-3.74	-6.20	-6.39	-1.87
PBE@FLO-SCAN	-8.28	-7.02	-5.21	-11.95	-4.42	-6.84	-8.11	-3.48
FLO-PBE	6.03	-10.36	2.02	3.14	1.12	-2.19	2.29	-12.09
ΔE^{SIC}	13.20	-4.26	6.02	13.86	4.86	4.01	8.68	-10.22
ΔE_{SB}^{SIC}	9.70	4.94	5.82	11.34	1.50	4.32	7.28	-3.14
SCAN	-7.91	-7.75	-7.74	-11.09	-6.63	-6.13	-9.39	-11.61
SCAN@FLO	-6.05	-6.21	-6.37	-9.27	-6.22	-5.50	-5.66	-8.52
FLO-SCAN	4.56	-7.01	-0.72	1.69	-2.32	-2.21	2.42	-14.99
ΔE^{SIC}	10.61	-0.8	5.64	10.96	3.90	3.29	8.08	-6.47
ΔE_{SB}^{SIC}	7.54	3.52	4.05	8.65	1.09	3.14	5.84	-2.50
Reference 43	6.3	19.5	10.9	13.2	3.9	17.2	1.6	33.8

SIC and density-related error

- ❑ How much of the DFA barrier improvement is due to using FLO density?
- ❑ In spirit of density-corrected DFT*

Forward barrier in PZSIC

DFA part evaluated on
PZSIC density

$$\Delta E_f = E(TS) - E(R) = \Delta E^{DFA@FLO} + \Delta E^{SIC}$$

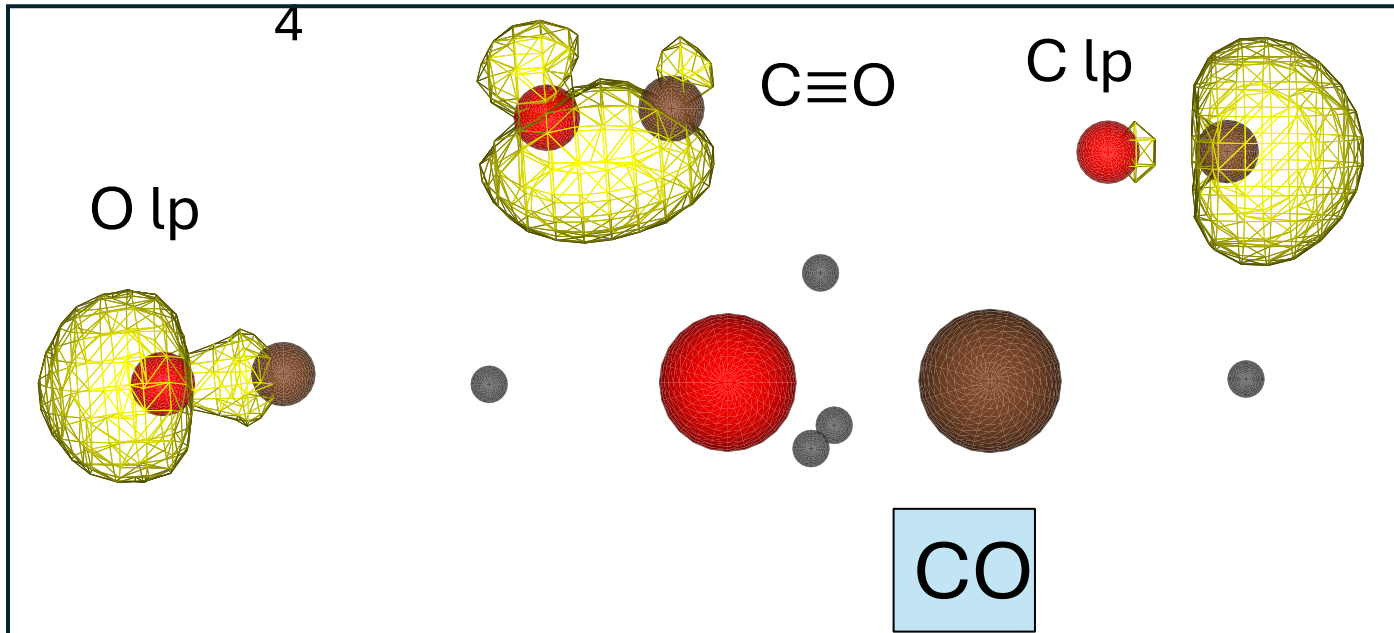
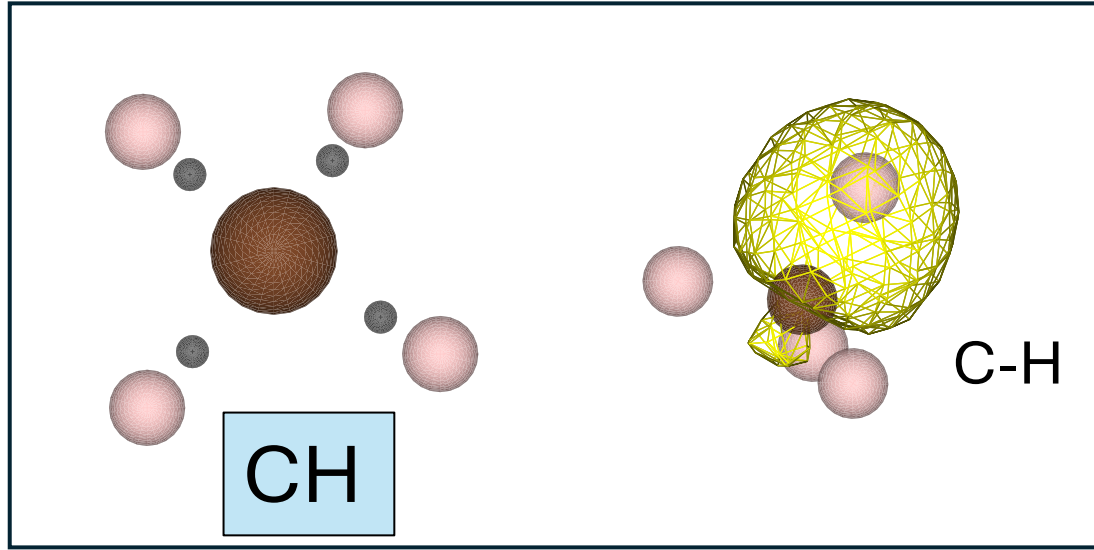
$$\Delta E^{DFA@FLO} = E^{DFA}(TS)[n_{FLO-DFA}] - E^{DFA}(R)[n_{FLO-DFA}]$$

$$\Delta E^{DFA} - \Delta E^{DFA@FLO}$$

Density-related part of SIC
correction

*Sim, Song, and Burke, J. Phys. Chem. Lett. **9**, 6385 (2017)

Optimized FODs $\leftrightarrow$ optimized FLOs



FODs and chemistry:

- FOD along single bonds $\leftrightarrow$ hybrid bond orbitals
- Pairs, triangles of FODs along multiple bonds $\leftrightarrow$ banana bond orbitals, etc.
- Lone pairs
- Good starting points for FOD optimization